Changing sun direction without breaking the building

Mehmet Ozturk. Task report for MyArchitectAI. Tests run on 6 November 2026.

Bottom line

No tool I tested moves the sun cleanly and also keeps the scene intact. The closest is the Sun-Direction LoRA on exteriors. I would pilot its Apache-2.0 4B version on exteriors only, behind automatic checks, and would not ship interiors yet.

What I found

Options fall into three groups: LoRAs trained for lighting, general edit models, and commercial apps. I found only two built directly for sun position, and several public demos were paused or crashing. The full list is on page 2.

What I tried

I ran three tools on three real photos, then repeated the best one on my own GPU in its license-safe version. Space times include queue and upload. Cost was $0.

Tool	Sun moved?	Scene kept?	Time per image
Sun-Direction 9B (Space)	Yes on exteriors	Framing kept. Colors turned gray	20.5 to 26 s
Sun-Direction 4B + color prompt (my RTX 4070)	Yes on exteriors	Color mostly back. Some materials changed, objects added	81 to 103 s*
Qwen 3D Lighting (Space)	Partly. Old shadows stay	Reframed 16 to 19%. Added flower pots	6.5 to 17 s
FLUX.1 Kontext (Space)	No on exteriors	Best structure, because little changed	31.5 to 45 s

*Partly offloaded to system RAM, so this is not production speed.

What went wrong

- **Interiors:** every tool failed differently. Sun-Direction changed the sofa color, the table and the rug size. Kontext replaced the outside view. Qwen put a sun patch on the window wall, which is physically impossible.
- **Added content:** models invented flower pots, clouds, a tree and a ceiling light.
- **My first score misled me:** Kontext scored best because it barely changed the light, and the overall score missed the furniture changes. I also blamed the wrong step for the gray colors at first.
- **Light transfer:** applying only the model's light to the original kept every object, but old hard shadows stayed and halos appeared.

How I decided which result is better

I did not rank by one number. Each result had to pass three checks: the light really changed, the structure survived after correcting scale and shift, and individual objects survived in before and after crops. The numbers narrow the choice, but the final call is visual.

What I would put in production

Self-hosted serverless GPU running the 4B LoRA with a color prompt, then color correction, automatic structure checks with retry, and 4K upscaling. **Estimated cost:** about $0.01 per image, or $0.02 with one retry, using RunPod's RTX 4090 at $1.34 per hour. **Estimated speed:** 20 to 26 s, based on 9B timings. Neither figure was measured for 4B on that GPU. Hosted Qwen and Kontext APIs cost $0.02 to $0.05 per image, but their open versions failed this task in my tests. An always-on GPU costs about $650 a month and only makes sense above roughly 16,000 images a month.

My next step would be to test 4B on at least 30 exteriors and measure how often a result passes the checks. This was one seed on three images, so these are failure patterns, not reliable rates.

What is on the market

Option	Type	One line
Sun-Direction LoRA (FLUX.2 Klein)	Open LoRA	Trained for sun rotation and elevation on exteriors, using a rendered light ball. 4B is Apache-2.0; the 9B base is non-commercial.
Qwen-Edit Multi-Angle Lighting	Open LoRA	Eight directions plus above and below, by text. Author says some angles are uncontrolled.
Qwen-Edit Relight, Light Migration	Open LoRA	Relight from text, or copy the lighting of a reference image.
FLUX.1 Kontext	Open weights, API	General text editing. Dev weights are non-commercial; Pro is $0.04 per image on fal.
LTX-2.3 Relight IC-LoRA	Open LoRA, video	Sun direction and hardness for exterior video only.
IC-Light	Open model	Older SD 1.5 relighter with conflicting license labels.
Nano Banana family	Commercial API	Prompt-based editing, from $0.0336 per 1K image.
fal relighting	Commercial API	$0.04 per image. Underlying model not confirmed.
Magnific, Krea, Clipdrop	Commercial apps	Subscriptions. No evidence of architecture-specific relighting found.
LBM Relighting, UniRelight	Ruled out	LBM relights a foreground object and is non-commercial. UniRelight is non-commercial video.

Measured results

Structure is local correlation with the original after correcting scale and shift, from 0 to 1. Scale is the correction needed; 1.00 means the framing held. Light is the spatial change in brightness. A high structure score alone means little if the light barely changed.

Image	Tool	Structure	Scale	Light (EV)
Villa	Sun-Direction 9B (Space)	0.33	0.97	0.37
Villa	Qwen 3D Lighting (Space)	0.43	0.81	0.42
Villa	FLUX.1 Kontext (Space)	0.95	1.00	0.24
Villa	Sun-Direction 4B, no prompt (RTX 4070)	0.29	0.97	0.30
Villa	Sun-Direction 4B, color prompt (RTX 4070)	0.31	0.97	0.27
Concrete	Sun-Direction 9B (Space)	0.52	1.00	0.49
Concrete	Qwen 3D Lighting (Space)	0.60	0.84	0.41
Concrete	FLUX.1 Kontext (Space)	0.81	0.98	0.30
Concrete	Sun-Direction 4B, no prompt (RTX 4070)	0.52	0.99	0.50
Concrete	Sun-Direction 4B, color prompt (RTX 4070)	0.48	0.99	0.61
Living room	Sun-Direction 9B (Space)	0.40	0.99	0.40
Living room	Qwen 3D Lighting (Space)	0.60	0.82	0.50
Living room	FLUX.1 Kontext (Space)	0.65	1.00	0.37
Living room	Sun-Direction 4B, no prompt (RTX 4070)	0.48	0.99	0.29
Living room	Sun-Direction 4B, color prompt (RTX 4070)	0.49	0.99	0.31

Light transfer, which keeps the original pixels, scored 0.94 to 1.00 for structure but only 0.04 to 0.20 EV for light. Those images are on the next pages.

Before and after: Villa exterior



Original
Sun moved to the left of the image



Sun-Direction 9B (Space)
Structure 0.33, light 0.37 EV. Sun moved. Colors and pool turned gray.



Qwen 3D Lighting (Space)
Structure 0.43, light 0.42 EV. Haze on the left. Reframed, aspect changed.



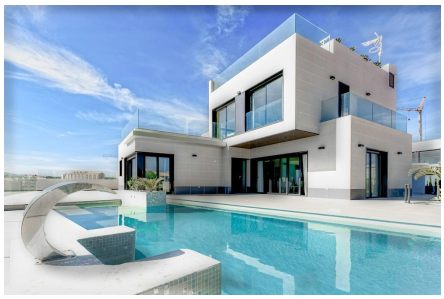
FLUX.1 Kontext (Space)
Structure 0.95, light 0.24 EV. Sun not moved. Slightly more saturated.



Sun-Direction 4B, no prompt (RTX 4070)
Structure 0.29, light 0.30 EV



Sun-Direction 4B, color prompt (RTX 4070)
Structure 0.31, light 0.27 EV. Best sun here. Steel sculpture looks like stone, facade warmer.



Light transfer from Sun 9B, smooth
Structure 0.97, light 0.14 EV



Light transfer from Sun 9B, sharp
Structure 0.95, light 0.15 EV



Light transfer from Qwen, smooth
Structure 0.99, light 0.07 EV

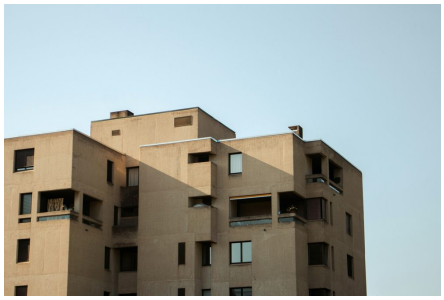


Light transfer from Kontext, smooth
Structure 1.00, light 0.10 EV



Intermediate overcast step, 4B
Input to the Sun-Direction pass.

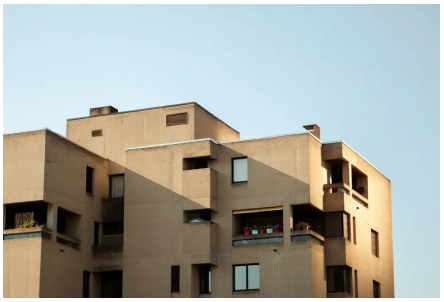
Before and after: Concrete apartment block



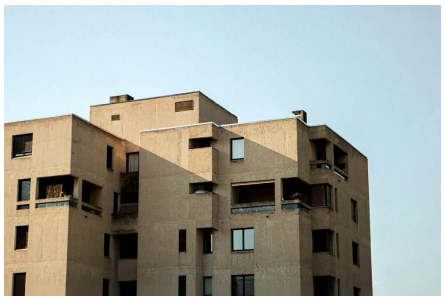
Original
Sun moved to the right of the image



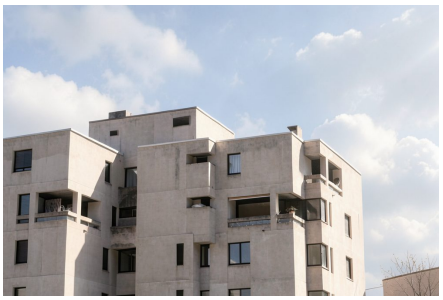
Sun-Direction 9B (Space)
Structure 0.52, light 0.49 EV. Sun moved. Beige concrete turned gray, sky pale.



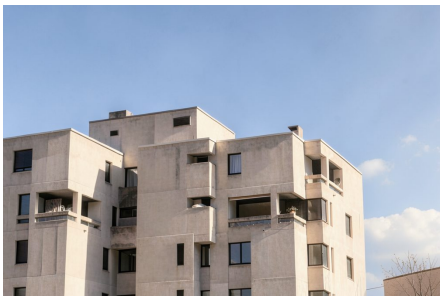
Qwen 3D Lighting (Space)
Structure 0.60, light 0.41 EV. Old shadow stays. Reframed. Added flower pots.



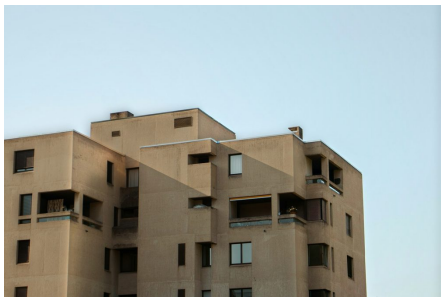
FLUX.1 Kontext (Space)
Structure 0.81, light 0.30 EV. Old shadow unchanged. Harsher texture, frame shifted.



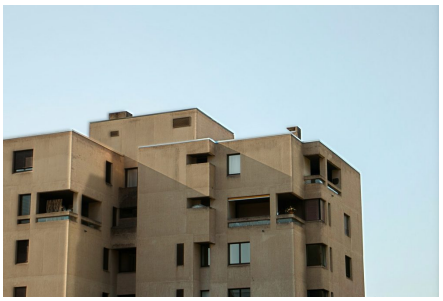
Sun-Direction 4B, no prompt (RTX 4070)
Structure 0.52, light 0.50 EV. Sun moved. Concrete faded, clouds and tree added.



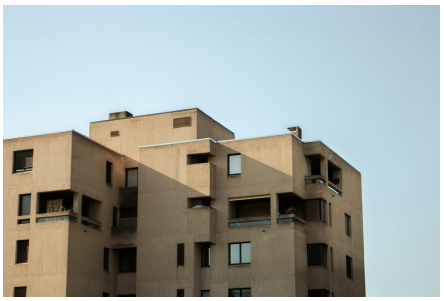
Sun-Direction 4B, color prompt (RTX 4070)
Structure 0.48, light 0.61 EV. Blue sky back, concrete paler. Tree still added.



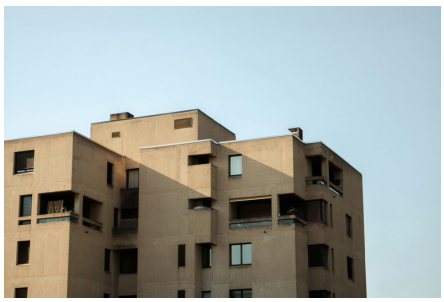
Light transfer from Sun 9B, smooth
Structure 0.98, light 0.20 EV. Objects intact. Old shadow stays, sun not moved.



Light transfer from Sun 9B, sharp
Structure 0.94, light 0.20 EV. Old shadow stays. Bright halo along the roof.



Light transfer from Qwen, smooth
Structure 0.99, light 0.09 EV



Light transfer from Kontext, smooth
Structure 1.00, light 0.04 EV



Intermediate overcast step, 4B
Input to the Sun-Direction pass.

Before and after: Living room



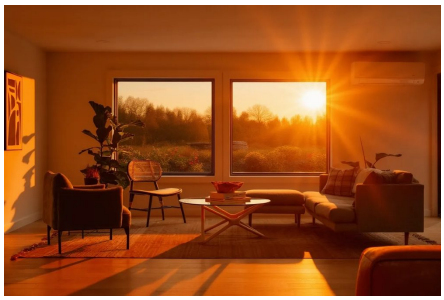
Original
Low sun from the back left, through the windows



Sun-Direction 9B (Space)
Structure 0.40, light 0.40 EV. Sofa, table, rug and view changed. Odd ring shadow.



Qwen 3D Lighting (Space)
Structure 0.60, light 0.50 EV. Objects kept. Reframed. Impossible sun patch on window wall.



FLUX.1 Kontext (Space)
Structure 0.65, light 0.37 EV. Orange cast, view replaced, sun on the wrong side.



Sun-Direction 4B, no prompt (RTX 4070)
Structure 0.48, light 0.29 EV



Sun-Direction 4B, color prompt (RTX 4070)
Structure 0.49, light 0.31 EV. Good light. Rug, armchair and view changed, extra ceiling light.



Light transfer from Sun 9B, smooth
Structure 0.96, light 0.15 EV. Objects intact. Light change is weak.



Light transfer from Sun 9B, sharp
Structure 0.94, light 0.15 EV. Objects intact. Halos and a ghost ring.



Light transfer from Qwen, smooth
Structure 1.00, light 0.04 EV



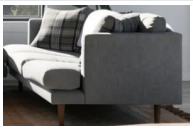
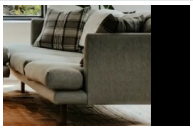
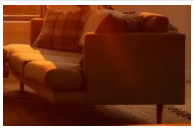
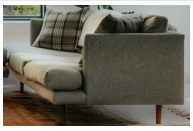
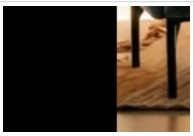
Light transfer from Kontext, smooth
Structure 0.98, light 0.16 EV



Intermediate overcast step, 4B
Input to the Sun-Direction pass.

Living room object check

Same crop in every column, after correcting each result's scale and shift. Black areas fell outside that tool's frame. The transfer column keeps the original pixels, but it also has the weakest light change.

	Original	Sun 9B	Sun 4B, color	Qwen	Kontext	Transfer
Armchair						
Sofa						
Coffee table						
Rug, left end						
Rug, front edge						

Appendix: problems log

Tools and access

Problem	Effect	What I did
Two Qwen Relight Spaces paused or crashing, Multi-Angle Space crashing	First choices unusable	Switched to Qwen 3D Lighting Control
Second Sun-Direction Space crashing	Only one working mirror	Used the linoyts Space
Anonymous ZeroGPU limit after three attempts	Six planned runs failed	Logged in with a Hugging Face account
Saved Hugging Face token was invalid	Runs still failed after login	Diagnosed it, logged in again through the browser
Sun-Direction reserves 150 s per run	Color runs refused with 147 s left	Moved those tests to my RTX 4070
Klein 9B is too large for a 12 GB card	Could not repeat the Space run locally	Used 4B, which is also the license-safe one
Local 4B needs offloading to system RAM	81 to 116 s per image	Recorded as a local test, not production speed
Sun-Direction needs a ball image made in the browser	No direct API call	Recreated the ball scene in Python
Openverse rate-limited image search	Partial search results	Used Unsplash instead
OpenAI pricing refused access, BFL prices came back blank	No first-party prices for those	Left out of cost figures
Google pricing page read only partly	Figures less certain	Marked as checked from a summary

Decisions that changed

First position	What changed
LBM was a candidate	Dropped: it relights a foreground object and is non-commercial
Free API credits were expected	None shown on Google, fal or Replicate
A render was one of the three test images	Replaced with a real photo, so renders remain untested
The villa was chosen for hard shadows	Its shadows are softer than expected and it is heavily processed
Sun-Direction looked like the clear winner	Changed after the interior object check
Gray colors seemed to come from the overcast pass	Saved that image. Color survived it, so the LoRA pass is the cause
The 9B model seemed the one to recommend	Replaced by 4B because 9B is non-commercial
Qwen reframing might have been my size settings	Same result at the Space's own sizes
Light transfer might solve everything	Objects were kept, but the sun did not really move on exteriors
Smooth transfer was the first setting	Light barely changed. The sharper setting added halos

Measurement problems

Problem	Fix or status
Structure score fell near zero on reframed images	Added scale and shift alignment
Kontext scored best while barely changing the light	Score light change separately
Score could not separate a correct shadow from an added object	Read the difference map by eye
Overall score missed the sofa, table and rug changes	Added per-object crops and measurements
Color check ignores brightness, so dark to light gray scores low	Not fixed. Crops are checked by eye
Position check gave a 1 px table shift while its shape had changed	Use object structure and crops

Photo credits, Unsplash: framesforyourheart, Clay Banks, Valentin Gerardi. All attempts are kept in the project's sonuclar folder, including a repeat Qwen run at the Space's sizes that gave the same result.